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The Weights of the Parts of the Brain in Several Species of Animals

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The relationship of brain weight to body weight and the expression of this relationship by "cephalization" formulae have received considerable attention. It is said that Tiedeman in 1836 was the first to study the weight of the encephalon in terms of body weight. Others who have continued this study are: Dubois (1897), von Bonin (1937), Quiring (1941), Count (1947), Sholl (1948) and Wirz (1950), to list but a few who have participated in this interesting study.

Probably the first quantitative study of the parts of the brain was that of Marshall (1892). Parts of the brain of the albino rat were studied quantitatively by Stewart (1918) and later Donaldson (1924) summarized the then known data on the weights of the brain and its parts, as well as the other parts and organs of the body, in the albino rat. Quantitative studies of the subdivisions of the brain of other vertebrates have been made by Larsell and von Berthelsdorf (1941), Sutter (1943), Wirz (1950) and Riese and Riese (1952). Unfortunately, all of these workers have not used exactly the same method of dividing the brain into its parts, and this limits our comparisons of the parts of the brain. A study of a series of adult animal brains, following the same method of dissection and made by the same person, has been published (Latimer, 1924, 1925, 1938b, 1942, 1946, 1950 and Latimer and Sawin, 1955). Much of the earlier quantitative literature on the human brain has been reviewed by Marshall (1892), Donaldson (1909), and Anthony (1928) and no attempt will be made to discuss this literature nor the quantitative studies on the parts of the developing brain, either in man or in various animals, for both the pre- and post-natal periods.

Materials and Methods

In an earlier report (Latimer, 1950) the four divisions of the brain, namely, prosencephalon, mesencephalon, cerebellum and medulla, all as percentages of the entire brain weight, are shown in bar graphs, arranged according to the increasing relative size of the prosencephalon. In this present study, the sums of the percentage weights of the prosencephalon and cerebellum, or the two suprasegmental parts, are represented by the unshaded areas in the bar graphs in Figure 1 and the sums of the two

parts of the brain stem, mesencephalon and medulla, are shown as the crosshatched areas for 10 species of animals including man. These bar graphs are arranged according to the increasing relative weights of the two suprasegmental parts, or prosencephalon and cerebellum, and the order in this figure is slightly different from that in the earlier figure. The turtle (Figure 1) comes between the chicken and the rat, while in the earlier

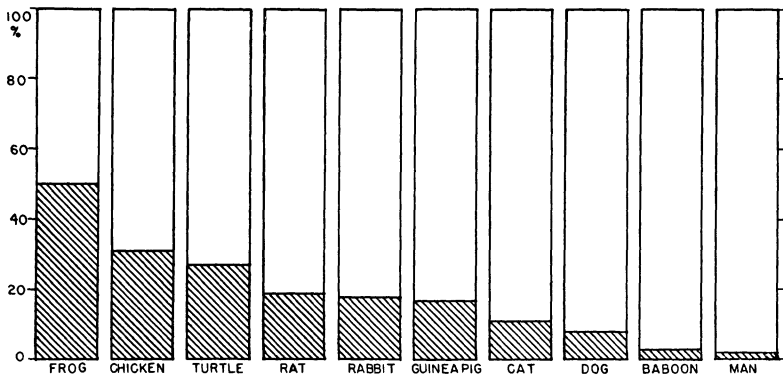


Fig. 1. A bar graph showing the percentages of total brain weight of the suprasegmental parts (prosencephalon and cerebellum) in the unshaded part of each column, and the brain stem (mesencephalon and medulla) in the crosshatched lower part of each column. The percentage values for each of these areas are given in Table 1.

figure it came between the rabbit and guinea pig. This present order seems to be much better. These percentage values for the two subdivisions of the brain are given in Table 1.

By employing this present method of studying the parts of the brain, the relative weights of the basic parts of the brain or the brain stem, with its primitive body functions, can be more easily compared with the suprasegmental parts of the brain, or with those parts of the brain containing the higher centers. This method shows more clearly the increase in the parts containing these higher centers in this series of animals. The data in Figure 1 and in all the tables include weights of the parts of male brains only for the turtle, rat, rabbit, guinea pig, cat and dog, although the relative proportions of the brain and its parts in the two sexes of the cat, dog and rabbit have been shown to be much alike, with only the mesencephalon having a significant sex difference in the cat and rabbit and only the medulla of the dog showing any sex difference. The uniform method used in dissecting all of the brains from this laboratory has been described in an earlier report (Latimer, 1938a) and will not be repeated. All but three of the brains in this collection, namely the rat, baboon and

man, were dissected and weighed by one person and the references for these three brains are given in Table 1.

Although quantitative measurements of the parts of the brain do not reveal the increasing complexity of structure upon which "intelligence" is very largely dependent, it seems reasonable to assume that the changes in the relative proportions of these two major divisions of the brain will show the phylogenetic development of the brain far better than merely the total weight of the entire encephalon. This method will not reveal finer degrees of organization of the central nervous system and differences between closely related groups of animals but it should show some of the main differences between the larger groups of animals.

Weights and Percentage Weights

Table 1 and Figure 1 show the weights of the suprasegmental parts of the brain increasing from 50% of the total weight of the brain in the frog to 98% in man, with a corresponding relative decrease in the weight of the brain stem. Furthermore, there is a rather well marked grouping of these brains as shown by the relative proportions of the two divisions. The chicken and turtle have suprasegmental parts of the order of 70% of the encephalon; the rat, rabbit and guinea pig, which represent the rodents in this very meager series of animals, have suprasegmental parts all slightly over 80% of the total brain weight. Next we find the cat and dog with brains approximately 90% of whose weight is found in these suprasegmental parts. Last of all, the two primates with suprasegmental parts forming 97 and 98% of the total brain weight. While these two primates are set off from the other mammals, one would not care to assume that the difference between the baboon and man is measured by the one per cent difference in the masses of the suprasegmental

Table 1. Weights of the parts of the brain as percentages of entire brain weight.

	Prosencephalon and cerebellum	Brain stem
Frog	50.5	49.5
Chicken	68.8	31.2
Turtle	72.7	27.4
Rat*	80.71	19.29
Rabbit	82.03	17.97
Guinea pig	83.25	16.75
Cat	89.18	10.64
Dog	92.14	7.86
Baboon*	96.7	2.4
Man*	97.87	1.95

* In this and in the two following tables, data for the rat are taken from Donaldson (1924), the baboon, from Riese and Riese (1952) and for man, from Donaldson (1909).

parts of their brains. This illustrates the statement that intelligence cannot be measured entirely by the weight of the brain nor by the weight

of the suprasegmental parts of the brain. Other details of structure and organization must be used in separating these two primate brains.

Thus it appears that the rodents have a different type of brain pattern, as far as quantitative measurements are concerned, from the cat and dog, and the primates have still a different pattern of brain proportions. There appears to be a specific pattern or plan in the proportions of the masses of the parts of the brain for each of the larger groups of animals, if such a conclusion may be drawn from so small a series of animals.

Table 2 shows the changes in the weights of the prosencephalon and the cerebellum with reference to the weights of the brain stem. The first two columns contain the weights in grams of the brain stem and the forebrain, and the third column, the ratios of the weights of the forebrain to the brain stem. These ratios increase from 0.98 times in the frog to 43 times in the human brain; thus while the forebrain of the frog is nearly the same size as the brain stem, the prosencephalon of man is 43 times the weight of the brain stem. Although each of these ratios is larger than its immediate predecessor, the changes are not uniform. The smallest difference (0.16) is found between the rat and the rabbit, and the largest change (35.8) is found between the ratios of the dog and the baboon. The difference between the ratios of the baboon and man is 6.76 or the second largest change in these ratios.

Table 2. Weights in grams of the brain stem, prosencephalon and cerebellum and the ratios of the suprasegmental parts to the brain stem.

	Brain stem grams	Prosencephalon grams	Ratios	Cerebellum grams	Ratios
Frog	0.049	0.048	0.98		
Chicken	1.11	1.83	1.65	0.460	0.41
Turtle	0.197	0.491	2.49	0.038	0.19
Rat	0.337	1.198	3.56	0.244	0.72
Rabbit	1.642	6.118	3.72	1.381	0.84
Guinea pig	0.671	2.803	4.18	0.532	0.79
Cat	2.98	20.61	6.92	3.96	1.38
Dog	6.03	63.52	10.53	7.37	1.22
Baboon	3.91	142.14	36.35	15.49	3.96
Man	27.	1164.	43.11	144.	5.33

The second column from the right (Table 2) shows the weights of the cerebellum in grams and the last column, the ratios of the weights of the cerebellum to the brain stem (column 1). These ratios of the cerebellum to the brain stem increase a total of 13 times, while the similar ratios of forebrain to brain stem increase 26 times, or twice as much as the cerebellar ratios. The ratio of the frog forebrain is omitted in obtaining this total increase, in order to obtain a more valid comparison with the cerebellar ratios, since the frog cerebellum was not weighed separately. There is more irregularity in the ratios of cerebellum to brain stem than in the ratios of forebrain to brain stem. Very likely the

amount and complexity of the activity of the animal is reflected in the relative size of the cerebellum and also in its ratio to brain stem. For example, the cerebellum of the sluggish turtle has a smaller ratio than that of the active chicken, likewise the guinea pig has a lower ratio than the very active rabbit, while the dog whose activities are confined to the ground has a slightly smaller cerebellum relative to brain stem than the cat with its skill on the ground and also in climbing. There are three animals with smaller ratios than the preceding animal in this ascending series, suggesting that the cerebellum in its relationship to the brain stem is modified by the type and amount of activity to which the animal is adapted. In general, the prosencephalon increases twice as fast compared to the brain stem as does the cerebellum, and the cerebellum increases 13 times as fast as the brain stem.

The weights of the brain stem and the combined weights of fore-brain and cerebellum, as percentages of body weight, are shown in the first two columns of Table 3 and the ratios of these percentage values, in the last column. The percentages of both the stem and the suprasegmental parts show a good deal of variation but the ratios of the percentage

Table 3. Brain stem and suprasegmental parts of the brain as percentages of body weight and the ratios of these percentages.

	Brain Stem % of body weight	Prosencephalon and cerebellum % of body wgt.	Ratios
Frog	0.105	0.107	1.02
Chicken	0.046	0.103	2.23
Turtle	0.023	0.063	2.74
Rat	0.235	1.004	4.27
Rabbit	0.088	0.401	4.56
Guinea pig	0.103	0.514	4.99
Cat	0.106	0.871	8.22
Dog	0.055	0.644	11.71
Baboon	0.0174	0.702	40.34
Man	0.056	2.713	48.45

weights of the combined percentages of the forebrain and cerebellum to brain stem again show a progressive change indicating that the fore-brain and cerebellum become progressively larger than the brain stem when these weights are compared to body weight, for these ratios increase from 1.02 for the frog brain to over 48 for the human brain. As in the preceding table, there appears to be a grouping of these ratios. The ratios for the three rodents are very much alike, all between four and five times, followed by a larger difference of approximately three, between the guinea pig and the cat and about the same difference between the cat and the dog. Again the greatest difference is between the dog and the baboon, a difference of over 28 and a difference of but a little over 8 between the baboon and man. In this table as in Table 2, the primates are set off very distinctly from the lower mammals with less difference

between the baboon and man than between the baboon and the dog. All of these ratios suggest that there is a type of brain, as far as masses of the parts are concerned, which is characteristic of the lower mammals and a distinct type in the proportionate masses of the brain parts in the primates. Thus the baboon brain, quantitatively, is more like the human brain than the dog type of brain. Can it be possible that the dog through long years of association with man is more intelligent than the primate baboon which has never had the chance to develop his potentially greater brain through association with man as has the dog?

There are three animals with suprasegmental parts larger, as percentages of body weight, than that of the baboon; namely, the cat, rat and man. The human forebrain and cerebellum taken together are nearly four times the percentage weight of the same parts in the baboon, and much greater than in any of the other animals listed in this table. The human brain is very distinctly demarked from the baboon brain in these percentage weights of the suprasegmental parts.

Figure 1 and all of the tables show that the suprasegmental parts of the brain increase more than the brain stem in this very small series of vertebrates. These changes in the size of the brain stem, with its basic or primitive centers as compared to the greater changes in size of the higher parts, or the suprasegmental parts of the brain, is shown in the weights of these two divisions as percentages of total brain weight (Figure 1 and Table 1), in the weights of each of these divisions in grams (Table 2) and in the weights of these two divisions as percentages of total body weight in Table 3.

Discussion

The brain is not a simple organ with a single function or at best a very limited number of functions, but rather it is composed of many closely interrelated cell centers, each with multiple connections. Furthermore, many of these pathways from the higher centers extend through the brain stem and pass into the spinal cord to connect the brain with the periphery of the body. Therefore the brain stem, as well as the cord, reflect to a degree the size and complexity of the higher centers, thus making it difficult to separate completely, by gross quantitative methods, the parts containing the higher centers from the basic or more primitive parts of the brain.

The area and weight of the cerebral cortex in man and in some animals has been studied by Kappers (1926) and included is a rather extensive review of the literature upon this subject. In the determination of the volume of the cortex, several methods have been employed resulting in

somewhat different estimated volumes. The surgical removal of parts of the brain, followed by a careful study of the behavioral pattern of these operated animals compared to normals, has been done by Lashley (1929 and 1948). Another method of investigating the quantitative anatomy of parts of the brain has been followed by Harman and Carpenter (1950) by determining the volumes of the basal ganglia of several primates including the human brain.

The size of the cerebellum in man and in animals has been studied by Kappers (1927), and the weights of the human cerebellum published by Mall (1909) have been studied statistically by Latimer (1946). Kappers gives percentage weights of the cerebellum of *Lepus cuniculus* as 15% of the total brain weight, which is very similar to the 15.13% for male and 15.37 for female rabbits (Latimer and Sawin, 1955). The percentage of 13.66 for the cerebellum of the domestic cat listed by Kappers is also much like the 14.39% for the cerebellum of the male cat (Latimer, 1938b). The relative weight of the cerebellum of 8.9% in the dog of unspecified sex reported by Kappers (1927) is much like the 9.60 and 8.75% of the total brain weight for the male and female dog cerebellum (Latimer, 1942). These data assembled by Kappers (1927) are the most extensive known data on the weights of the adult cerebellum in man and in animals. Larsell and von Berthelsdorf (1941) have studied the relationship between various lobules of the cerebellum and the related muscle masses in several species of animals.

Portmann (1942) and Sutter (1943) have studied the divisions of the avian brain, but they have used a different method of dividing the brain into its parts. They have separated the telencephalon from the diencephalon, after removing the cerebellum and corpora bigemina, thus leaving as the "stammrest" the diencephalon, the mesencephalon, without the corpora bigemina, and the medulla. Wirz (1950) has employed the same method of dividing the brain, and he has used these parts of the brain, rather than the weights of the entire brain, in studying the problem of cephalization. This method of dividing the brain used by Portmann, Sutter and Wirz, all working in the same laboratory, differs from the present method in grouping the diencephalon with the brain stem rather than with the telecephalon, with which it seems to be closely related in structure and in function. It is a question, as Wirz admits, in which group the diencephalon should be included. The thalamus is so closely related to the telencephalon, at least in man, that it has seemed best to group it with the cerebrum and cerebellum. The corpora bigemina of birds are the chief optic centers, and should be grouped with the cerebrum and cerebellum, but in the higher animals the chief optic

centers are found in the hemispheres. There are indeed problems as to where the divisions should be made when sectioning the brain, since all of the parts are so intimately and physiologically connected, at least in the higher mammals.

The quantitative as well as the morphological study of the brain has been a challenging subject over a very long period of time and there have been many attempts to classify animals by the size of the brain and its quantitative relationship to body weight, as seen in the use of the cephalization formulae. If there were as many quantitative data on the weights of the parts of the brain as there are for the weights of the entire encephalon, we might now have a clearer understanding of its relationship to phylogenetic development. Finer degrees of intelligence will probably not be detected by quantitative measurements of either the entire brain or of its parts, but the larger trends in phylogenetic development of the encephalon may be clarified by a more exhaustive study of the size of the various parts or divisions of the encephalon.

Summary

The suprasegmental parts of the brain, namely prosencephalon and cerebellum, range from 50% of the total brain weight in the frog to 98% in man. The brain stem including the medula and mesencephalon shows a corresponding decrease.

The weights in grams of the brain stem and the forebrain are approximately the same in the frog but the human forebrain is 43 times the weight of the human brain stem. The cerebellum shows an increase relative to the brain stem, but not to the same extent as the forebrain, for the ratio of the cerebellum to brain stem increases about 13 times, while the forebrain increases about 26 times.

The ratios of the suprasegmental parts to brain stem, with both of these divisions expressed as percentages of the body weight, increase from about 1 to 48 times.

The weights of the brain stem and the suprasegmental parts vary a good deal, but the ratios of the suprasegmental parts to the brain stem display in general a progressive increase, but not a uniform increase for there appears to be a grouping of these ratios, suggesting four brain types: first, the chicken and turtle; second, a group of rodents composed of the rat, rabbit, and guinea pig; third, the two higher mammals or the cat and dog with a rather wide range of ratios, and fourth the two primates, the baboon and man. This series is by no means sufficiently extensive to more than suggest these trends in the weights and relative weights of these two divisions of the brain. Many more species would

be needed before final or even tentative conclusions would be justified. This study of the small series merely suggests that the larger groups of animals have specific types of brains, judged by the quantitative differences in the weights of their parts.

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